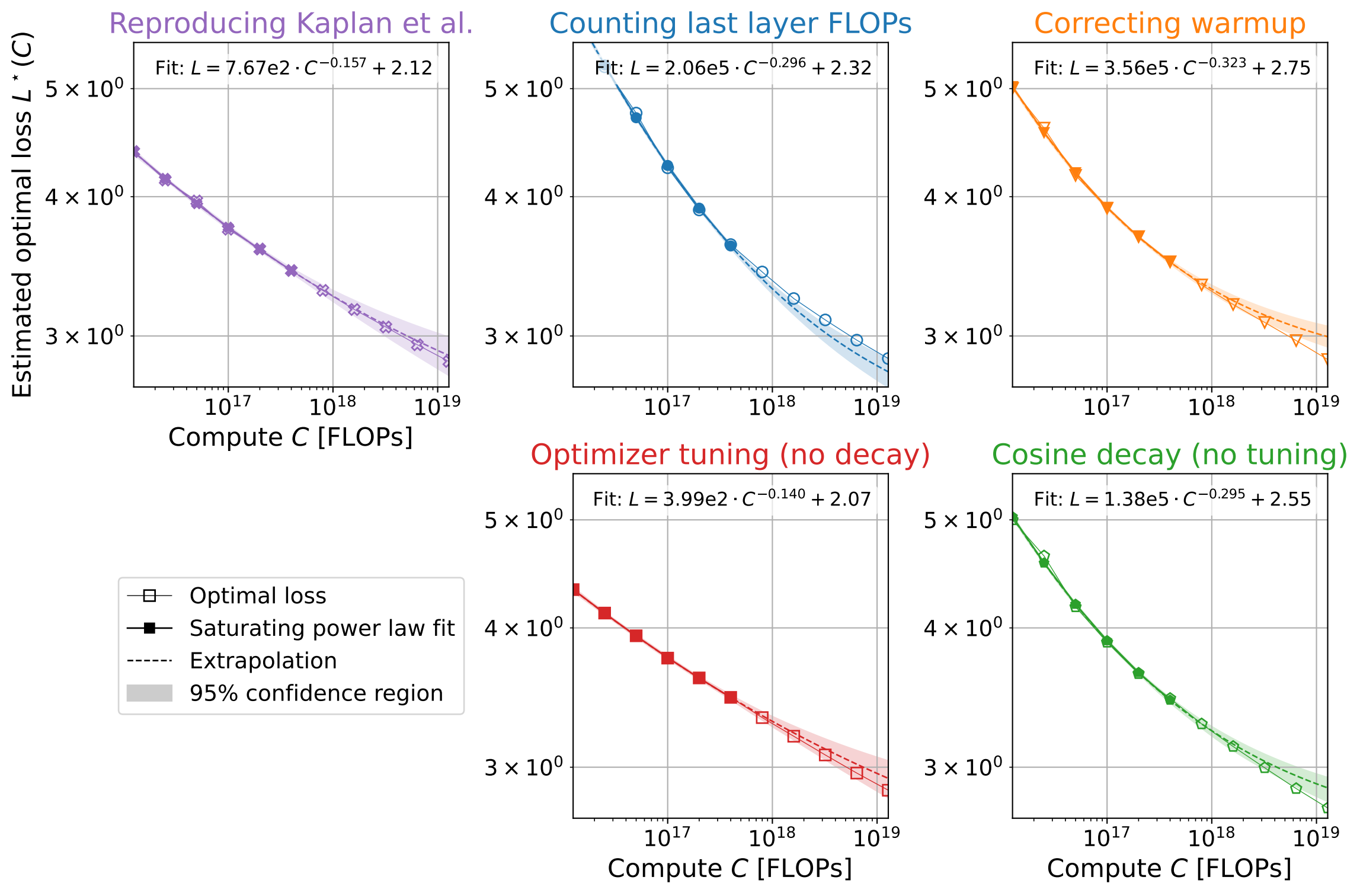
Reproducing Kaplan et al.
Fit: $L = 7.67e2 \cdot C^{-0.157} + 2.12$
Counting last layer FLOPs
Fit: $L = 2.06e5 \cdot C^{-0.296} + 2.32$
Correcting warmup
Fit: $L = 3.56e5 \cdot C^{-0.323} + 2.75$
Optimizer tuning (no decay)
Fit: $L = 3.99e2 \cdot C^{-0.140} + 2.07$
Cosine decay (no tuning)
Fit: $L = 1.38e5 \cdot C^{-0.295} + 2.55$
Estimated optimal loss $L^\star(C)$
Compute $C$ [FLOPs]
$5 \times 10^0$
$4 \times 10^0$
$3 \times 10^0$
$10^{17}$
$10^{18}$
$10^{19}$
Optimal loss
Saturating power law fit
Extrapolation
95% confidence region